

Web Accessibility

A practical guide to
designing inclusive
digital experiences

What does web accessibility mean?

Definition:

Making the web usable for all, including people with disabilities (visual, auditory, motor, cognitive).

Importance:

- Promotes inclusivity
- Meets legal standards
- Enhances user experience universally

Key Aspects of Web Accessibility

Perceivable	Present information so users can perceive it (e.g., text alternatives, adaptable content, distinguishable elements).
Operable	Ensure all users can operate interface components (e.g., keyboard accessibility, adequate time for interaction, seizure-free content).
Understandable	Make content and operation clear and predictable (e.g., readable text, consistent functionality, error prevention).
Robust	Ensure content can be interpreted reliably by diverse technologies (e.g., adherence to web standards, compatibility with assistive technologies).

“Edge Users”

People with disabilities were once considered "edge users" in UX, with unique needs differing from the majority.

Includes people with disabilities, older adults, non-native language speakers, and users with low bandwidth or older technology.

1.3 billion people live
with disabilities

That's 16% of the global population
or 1 in 6 people

an opportunity to create
a more **inclusive design**
practice for yourself

WCAG Levels of Conformance

Level A	• Basic accessibility needs • Addresses fundamental barriers
Level AA	• Comprehensive accessibility, often meets legal requirements • Covers more barriers, wider audience
Level AAA	• Highest accessibility standard • Challenging to achieve for all content types

Semantic HTML

Importance:

Provides meaningful structure to web content, enabling assistive technologies to better interpret and navigate the site, thus improving the user experience for people with disabilities.

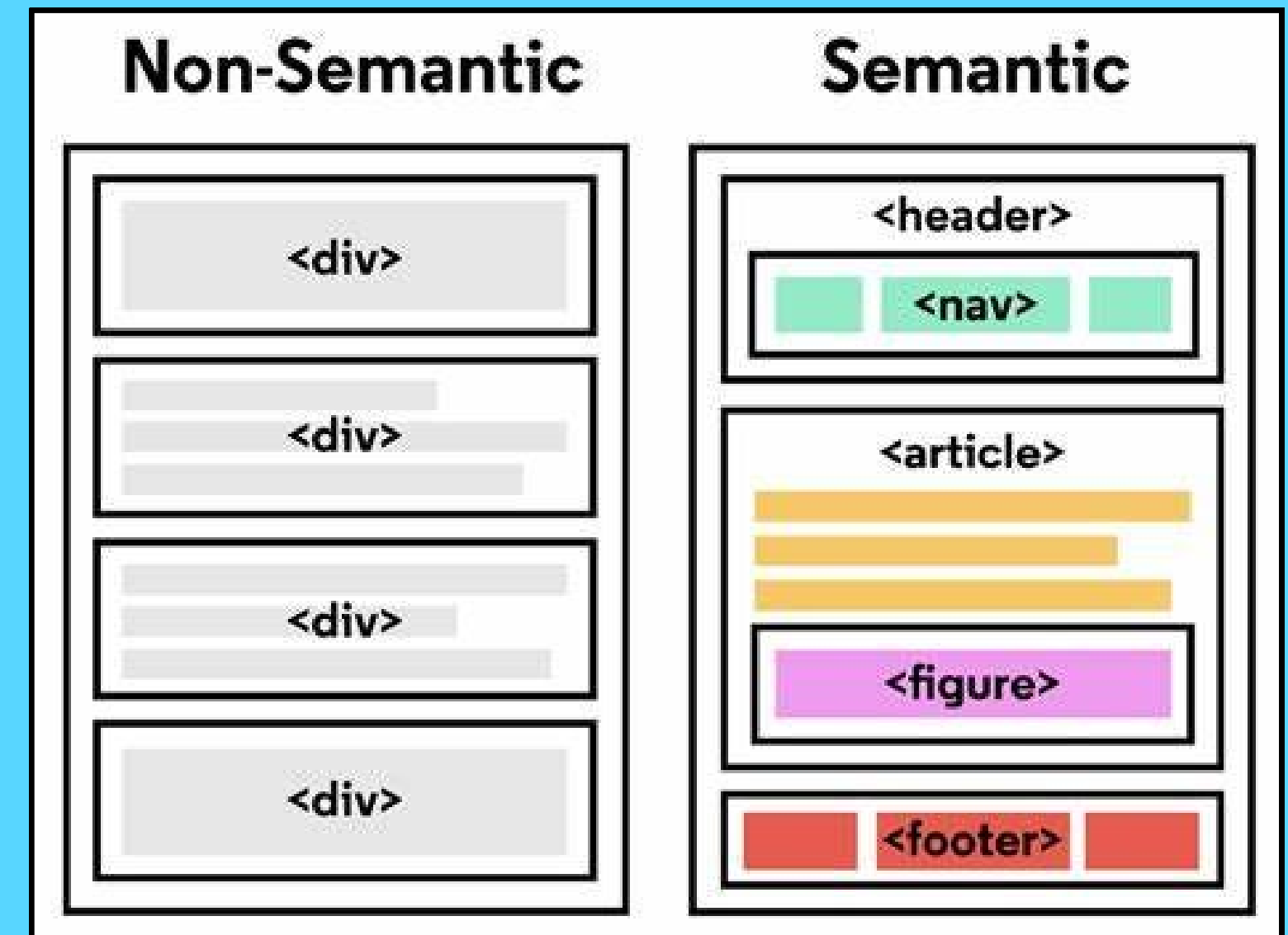
Best Practices:

- Use Appropriate HTML Elements (e.g., `<article>`, `<section>`, `<nav>`, `<footer>`) to provide clear meaning and structure.
- Ensure Proper Heading Hierarchy: Use headings (`<h1>` to `<h6>`) in a logical order to create a clear and accessible document outline for screen readers.

Semantic HTML

Categories of Common and Important Semantic Elements

- Document structure tags
- Textual meaning tags
- Media type tags
- Correlation tags



How screen readers work

Function:

Convert digital text to speech or Braille output.

Impact:

Facilitate independent access to digital information.

Capabilities:

- Interpret HTML and ARIA (Accessible Rich Internet Applications) attributes for contextual information.
- Support navigation via keyboard shortcuts or touch gestures.
- Provide feedback on interactive elements (buttons, forms).

Example screen reader workflow:

1. Starting the Screen Reader:

- The user launches the screen reader software (e.g., JAWS, NVDA, VoiceOver).
- The screen reader begins reading the content of the current window.

2. Navigating a Web Page:

- The user presses keyboard shortcuts to navigate by headings (e.g., 'H' key), links (e.g., 'K' key), or form elements (e.g., 'F' key).
- The screen reader announces each element and its context (e.g., "Heading level 1: Welcome to our site").

3. Interacting with Elements:

- When encountering a form, the user can navigate to input fields using the 'Tab' key.
- The screen reader announces each field (e.g., "Edit, First Name, required").
- The user fills out the form and submits it using keyboard commands.

Navigation

Importance:

- Consistent navigation helps users, including those with cognitive disabilities, find information easily and navigate the site without confusion.

Best Practices:

1. Consistent Menu Placement: Keep navigation menus in the same location on all pages.
2. Descriptive Links: Use clear and descriptive link text.
3. Skip Navigation Links: Provide a "skip to content" link for screen readers to bypass repetitive navigation.

Navigation

Importance:

Consistent navigation helps users, including those with cognitive disabilities, find information easily and navigate the site without confusion.

Best Practices:

- **Consistent Menu Placement:** Keep navigation menus in the same location on all pages.
- **Descriptive Links:** Use clear and descriptive link text.
- **Skip Navigation Links:** Provide a "skip to content" link for screen readers to bypass repetitive navigation.

Typography

Importance:

- Ensures readability for all users
- Poor typography can lead to user frustration and hinder comprehension.

Best Practices:

- Readable Fonts: Use simple, sans-serif fonts like Arial, Verdana, or Helvetica.
- Font Size: Ensure a minimum font size of 16px for body text.
- Line Height: Use sufficient line spacing (1.5 times the font size) to improve readability.
- Avoid All Caps: Text in all capital letters can be harder to read.

Typography

Atkinson Hyperlegible

Enhanced letterforms.
Open counters.
Exaggerated spurs.
Circular motifs.

Increased **recognition**.

B8 1Iil

B8 1Iil

Recognizable Footprints
Character boundaries clearly defined, ensuring understanding across the visual-ability spectrum

Differentiated Letterforms
Similar letter pairs differentiated from each other to dramatically increase legibility

QGEFpqirO0

Unambiguous Characters
Increase legibility for people with low vision or poor reading comprehension

ER79jr Csa36

Exaggerated Forms
Clarify potential misreadings

Opened Counterspace
Defines open spaces better

aGbgrpqu

Åö8ij%?i,:

Angled Spurs
Increase recognition and define distinctive style

Circular Details
Link to the history of Braille Institute and Braille dots

! " # \$ % & ' () * + , - .
/ 0 1 1 2 3 4 5 6 7 8 9 : ;
< = > ? @ A B C D E F G H I
J K L M N O P Q R S T U V W
X Y Z [\] ^ _ ` a b c d e
f g h i j k l m n o p q r s
t u v w x y z { | } ~ ¡ ¢ £
¤ ¥ ¦ § ¨ © ª « ¬ ® ¯ ° ± ²
³ ´ µ ¶ · ¸ ¹ º » ¼ ½ ¾ ¿ À
Á Â Ã Ä Å Æ Ç È É Ê Ë Ì Í Î
Ï Ð Ñ Ò Ó Ô Õ Ö × Ø Ù Ú Û Ü
Ý Þ ß à á â ã ä å æ ç è é ê
ë ì í î ï ð ñ ò ó ô õ ö ÷ ø
ù ú û ü ý þ ÿ ı Ł ł Œ œ Š š
Ÿ Ž ž ƒ ˆ ˇ ˘ ˙ ˚ ˛ ˜ ˝ π
– — ‘ ’ ‚ “ ” „ † ‡ • … ‰ ‹
› ⁄ € ℓ ™ Ω e ò Δ ∏ Σ − ⁄ ·
√ ∞ ∫ ≈ ≠ ≤ ≥ ◇ fi fl

Complete Character Set
246 glyphs per font

Atkinson
Hyperlegible

Regular / *Italic*
Bold / ***Bold Italic***

© Braille Institute of America, 2019

Available weights

Contrast

Importance:

- Proper contrast between text and background ensures that content is readable for users with visual impairments, including color blindness.

Best Practices:

- High Contrast Ratios: Use a contrast ratio of at least 4.5:1 for regular text and 3:1 for large text.
- Avoid Text on Images: If text is placed over an image, ensure there is sufficient contrast or use a text background.

Contrast Checker

[Home](#) > [Resources](#) > Contrast Checker

Foreground

Hex Value

#000000

Color Picker

Alpha

1

Lightness

Background

Hex Value

#58D6FF

Color Picker

Lightness

Contrast Ratio

12.47:1

[permalink](#)

Normal Text

WCAG AA: **Pass**

WCAG AAA: **Pass**

The five boxing wizards jump quickly.

Large Text

WCAG AA: **Pass**

WCAG AAA: **Pass**

The five boxing wizards jump quickly.

Graphical Objects and User Interface Components

WCAG AA: **Pass**

Text Input

Contrast Checker

[Home](#) > [Resources](#) > Contrast Checker

Foreground

Hex Value

#737373

Color Picker

Alpha

1

Lightness

Background

Hex Value

#05C1FF

Color Picker

Lightness

Contrast Ratio

2.27:1

[permalink](#)

Normal Text

WCAG AA: **Fail**

WCAG AAA: **Fail**

The five boxing wizards jump quickly.

Large Text

WCAG AA: **Fail**

WCAG AAA: **Fail**

The five boxing wizards jump quickly.

Graphical Objects and User Interface Components

WCAG AA: **Fail**

Text Input

Alt text

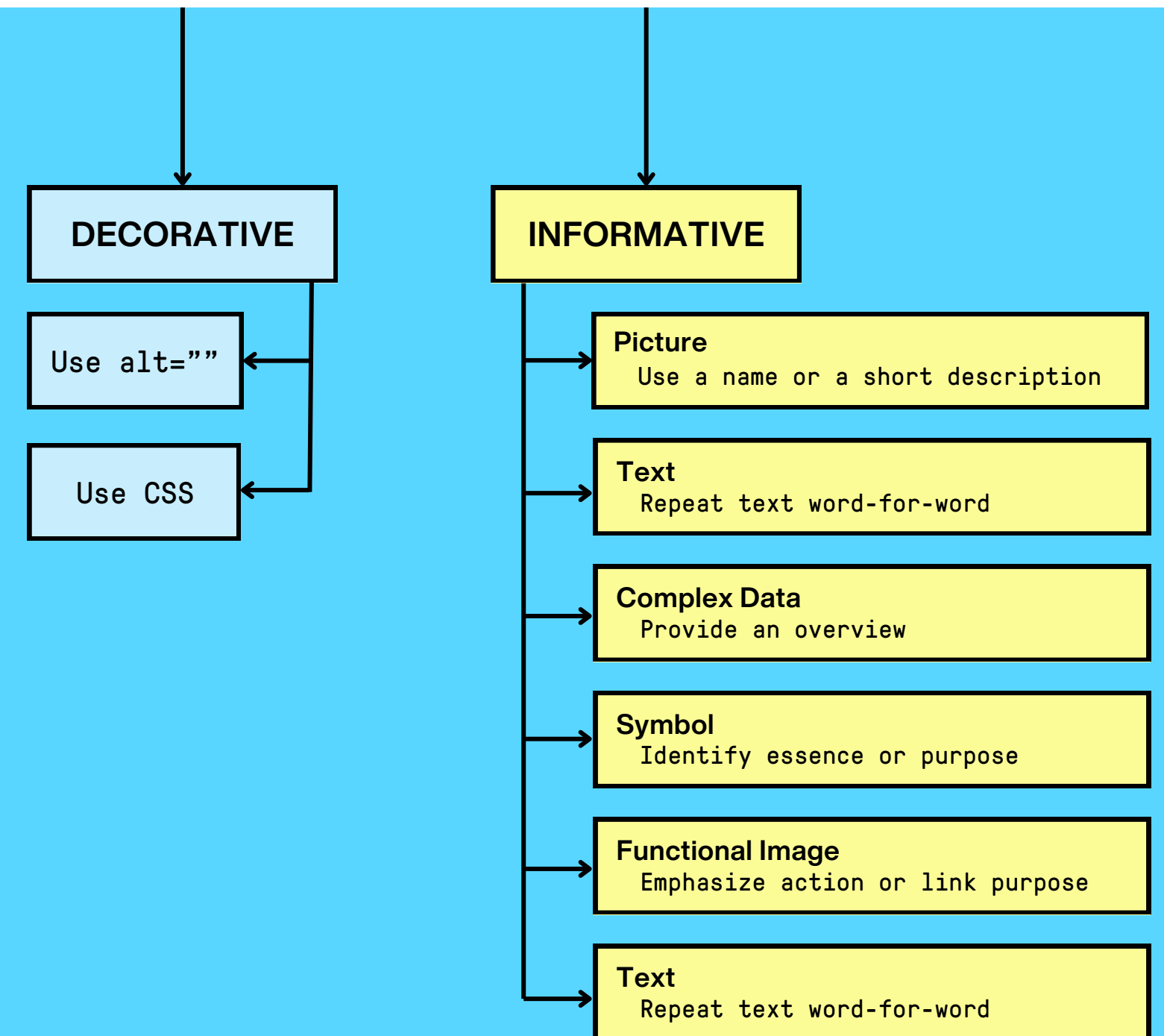
Importance:

Alt tags provide descriptive text for images, which is essential for screen readers and for search engine optimization (SEO).

Best Practices:

- Descriptive Alt Text: Clearly describe the content and function of the image.
- Use Null Alt Text for Decorative Images: Use `alt=""` for images that are purely decorative.

What is the purpose of the image?



Accessibility Testing Tools

- **WAVE:** Visual feedback tool by WebAIM.
- **Axe:** Browser extension for accessibility issues.
- **Lighthouse:** Auditing tool in Chrome DevTools.
- **Accessibility Insights:** Microsoft's tool for checks and assessments.

Accessibility Testing Tools

- **Keyboard-only Navigation:** Manual testing for keyboard access to interactive elements.
- **Code Validators**
 - W3C Markup Validation Service: Checks HTML/XHTML code.
 - W3C CSS Validator: Validates CSS code.

Accessibility Testing Tools

- **Chrome DevTools:** Accessibility inspection and testing.
- **Firefox Accessibility Inspector:** Tools for auditing and improving accessibility.

Accessibility Testing Tools

- **Contrast Checker by WebAIM:** Ensures text contrast.
- **Accessible Colors:** Finds accessible color combinations.

And most importantly...

Include people with
disabilities in usability
testing!

Emerging technology

AI-powered tools for automated testing, voice user interfaces like Alexa, AR/VR inclusivity features, and enhanced mobile accessibility (e.g., VoiceOver, TalkBack) are key trends.

Personalization options and accessible IoT devices further ensure usability for diverse needs.